



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)
SATHYAMANGALAM - 638 401

Approved Questions S2 Semester

22CH203 – ENGINEERING CHEMISTRY II

Q. No.	Questions
	Unit 1
1.	A laboratory constructs an electrochemical setup using zinc and copper metals placed in their respective ionic solutions. The zinc electrode gradually dissolves while copper ions get deposited as solid copper. A voltmeter connected between the electrodes shows a steady potential difference under standard conditions. The system operates without any external power supply. The experimenter observes directional electron flow through the external circuit. Analyze the system to determine the origin of the observed potential difference and infer how chemical reactions establish a driving force for electron movement across the circuit while maintaining charge balance within the system.
2.	A researcher compares two electrochemical cells, one operating under standard conditions and another with altered ion concentrations. The measured potential differs significantly between the two systems. Both systems use the same electrode materials but vary in solution composition. The researcher attempts to relate this observation to thermodynamic principles governing electrochemical reactions. Determine how variations in chemical conditions influence electrode potential and justify the thermodynamic basis for potential generation in electrochemical systems.
3.	An electrochemical system is represented using symbolic notation showing two half-cells separated by a salt bridge. One half-cell contains a metal immersed in its ions, while the other contains a different metal-ion pair. The notation includes arrows indicating reaction direction. A student misinterprets the sign of the cell potential and predicts incorrect electron flow direction. Interpret the correct representation of the electrochemical cell and infer how sign conventions determine the direction of electron flow and feasibility of the reaction.
4.	A student assembles a half-cell using a metal electrode dipped in an electrolyte but is unable to measure its absolute potential directly. The student then connects it to a standard hydrogen electrode to obtain a measurable value. The setup produces a stable voltage reading. Analyze why a reference electrode is necessary and determine how it enables measurement of electrode potential.
5.	In an experiment, a metal electrode is immersed in its ionic solution, and a potential develops at the interface. The system shows no external current flow but still maintains a measurable potential difference across the interface. Interpret the origin of this interfacial potential and infer the role of equilibrium in its formation.

6.	A closed electrochemical circuit shows continuous electron flow through a wire while ions move through the electrolyte. The system maintains neutrality despite ongoing reactions. Determine how ionic movement supports sustained electron flow in the circuit.
7.	Two different metals are connected without an electrolyte, and no current is observed despite a difference in their properties. Infer why an electrolyte is essential for current generation in electrochemical systems.
8.	A laboratory electrochemical cell is constructed using two electrodes immersed in electrolyte solutions of different ionic strengths, and the system is allowed to reach equilibrium under open-circuit conditions while maintaining constant temperature and pressure, after which the potential difference is measured and later compared with the voltage recorded when a small current is drawn through an external circuit, revealing a noticeable difference between the two readings. Analyze the observed difference between the measured potentials under open-circuit and load conditions, and determine the underlying electrochemical principles governing this variation while justifying the role of internal resistance and system equilibrium.
9.	An industrial electrochemical setup operates under varying environmental conditions where temperature increases gradually and reactant concentration changes due to continuous consumption, while pressure variations also occur in a gas electrode system, leading to fluctuations in measured cell potential over time. Determine how variations in concentration, temperature, and pressure influence the observed EMF and justify the combined effect of these parameters using electrochemical reasoning.
10.	A researcher evaluates two electrochemical cells, one designed to operate reversibly under controlled slow reactions and another subjected to rapid discharge with significant energy losses, and compares their efficiencies while attempting to relate electrical output to thermodynamic feasibility. Analyze the differences between the two cells in terms of reversibility and infer how these differences affect the relationship between EMF and Gibbs free energy.
11.	A standard hydrogen electrode is coupled with an unknown metal electrode under standard conditions, and the measured EMF is used to determine the relative position of the metal in an electrochemical series for predicting reaction feasibility. Interpret how the measured EMF can be used to position the metal in the electrochemical series and infer its reactivity.
12.	An electrochemical cell is analyzed under standard conditions where all reactants are maintained at unit activity, and the EMF value is recorded to compare it with values obtained under non-standard laboratory conditions. Determine the significance of standard EMF and justify its role in comparing electrochemical systems.
13.	A student measures EMF using a voltmeter directly across a working electrochemical cell that is simultaneously supplying current to a load, leading to inaccurate readings. Identify the limitation in the measurement approach and infer a more appropriate method for accurate EMF determination.

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14.	An electrochemical reaction shows a positive EMF value during operation, and the researcher attempts to infer the spontaneity and energy change associated with the reaction. Interpret the significance of positive EMF in relation to Gibbs free energy and reaction feasibility.
15.	A metal electrode is immersed in an aqueous electrolyte containing both cations and anions, where the metal surface acquires a net charge due to electron transfer. The nearby ions rearrange themselves forming a structured region near the interface. The compactness and distribution of ions vary depending on thermal motion and electrostatic attraction. Experimental observations indicate that the simple rigid-layer assumption fails to explain the diffuse nature of ion distribution at higher temperatures. Analyze how different theoretical models interpret the structure of the electrical double layer at the electrode–electrolyte interface and justify which model best reconciles both compact and diffuse charge distributions under varying conditions.
16.	An electrochemical cell shows variation in interfacial potential when the concentration of electrolyte changes. Measurements reveal that the potential drop is not linear but varies significantly close to the electrode surface. The electric field strength is highest near the electrode and gradually decreases into the bulk solution. Determine how the potential gradient develops across the electrical double layer and evaluate its relationship with charge distribution and capacitance at the interface.
17.	In a system involving polar solvents, it is observed that solvent molecules orient themselves near the electrode surface even before significant ion accumulation occurs. Additionally, certain ions are found to specifically adsorb onto the electrode, altering the effective surface charge. Interpret the combined influence of solvent orientation and specific ion adsorption on the effective structure and behavior of the electrical double layer.
18.	During an electrochemical reaction, the rate of electron transfer is observed to change when the electrolyte concentration is varied. The structure of the double layer shifts accordingly, influencing the accessibility of reacting species to the electrode surface. Analyze how changes in the electrical double layer influence electrode kinetics and reaction rates.
19.	An interface between a metal and electrolyte exhibits measurable changes in surface tension when an external potential is applied. These changes correlate with variations in surface charge density. Evaluate the relationship between surface charge density and electrocapillary effects at the interface.
20.	A system shows that increasing ionic strength reduces the thickness of the diffuse layer significantly. Infer how ionic strength affects the structure of the electrical double layer
21.	An electrode exhibits higher capacitance when placed in a concentrated electrolyte compared to a dilute one. Determine the reason for increased double layer capacitance in concentrated solutions

22.	A lithium-ion battery under high discharge is observed to show uneven ion distribution near the electrodes, where lithium ions accumulate on one side while depletion occurs on the other, despite the presence of an electric field driving ions across the electrolyte. The electrolyte viscosity is moderate, and no external stirring is applied, while temperature remains constant throughout the process. Analyze how the combined effects of migration and diffusion govern the observed ion distribution and determine the dominant transport mechanism under the given conditions while justifying your reasoning using electrochemical transport principles.
23.	An electrochemical cell using an aqueous electrolyte exhibits reduced efficiency when electrolyte concentration is increased beyond a certain level, even though higher ion availability is expected to enhance conductivity. Measurements reveal increased ion interactions and reduced mobility at higher concentrations. Evaluate the relationship between electrolyte concentration and conductivity and determine why increased concentration leads to reduced efficiency beyond an optimal point
24.	A fuel cell shows voltage drop during prolonged operation, where analysis indicates buildup of reaction products near the electrode surface, causing a local change in ion concentration and affecting charge transport. Determine how concentration polarization develops in this system and assess its impact on overall cell performance using transport principles.
25.	An electrolyte solution under an applied electric field shows that cations move faster than anions due to smaller size and higher mobility, contributing unevenly to current flow. Analyze how transport numbers arise in this system and determine their significance in current distribution.
26.	A battery exhibits increased internal resistance as it ages, even though electrode materials remain unchanged, and electrolyte conductivity slightly decreases over time. Evaluate the factors contributing to increased internal resistance and infer its effect on battery performance.
27.	An electrochemical system experiences enhanced ion transport when stirred externally, even though no changes are made to electric field or concentration. Determine the role of convection in this system and justify its effect on transport enhancement.
28.	In an electrolyte under steady conditions, both electric field and concentration gradient are present, influencing ion flux simultaneously. Interpret how the Nernst-Planck framework integrates these effects to govern ion movement.
29.	A laboratory team constructs an electrochemical setup using a zinc rod immersed in zinc sulfate solution and a copper rod immersed in copper sulfate solution, connected via a salt bridge containing potassium nitrate. During operation, they observe electron flow through the external circuit and ion migration through the salt bridge, but also note a gradual decrease in voltage over time due to changes in ion concentration near the junction. Analyze the working mechanism of this electrochemical cell by determining the role of each component and infer the reasons for voltage variation with respect to junction behavior and ion movement within the system.



Q. No.	
	Unit 1

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$$\frac{dy}{dx} - \frac{2y}{x} = 2x$$

17.	A signal y is given by $y = ae^{4x}$ Form the
18.	A robot moves with constant acceleration a and initial velocity u. Its displacement s is given by $y = mx + \frac{1}{2}at^2$ Form diff
19.	Fluid flow through a pipe is given by $y = \frac{a}{x} + b$ Form the

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34.	A motorb Formula
35.	In a cooli
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Approved Questions
S2 Semester**22AI206/22AM206/22CS206/22IT206 – DIGITAL COMPUTER ELECTRONICS**

Q. No.	Questions
	Unit 1
1.	A calculator device internally processes numbers in binary but must display outputs in a user-friendly format. The display has limited space, and binary representation appears lengthy. The system designer must choose between octal and hexadecimal formats for output representation while ensuring accuracy and minimal processing overhead. Determine the most appropriate number system for display and justify the decision based on efficiency and usability.
2.	A program converts binary values into decimal but fails when fractional values are present, producing incomplete results. The integer portion is processed correctly using positional weights, but the fractional portion is ignored. The developer must identify missing logic affecting correctness. Identify the error in the program and determine its effect on output accuracy.
3.	A telemetry system converts decimal data into binary and then into octal for transmission. Engineers observe incorrect outputs when fractional values are processed. The system uses standard conversion techniques but suffers from truncation and grouping inconsistencies, affecting final transmission accuracy. Diagnose the cause of incorrect outputs and infer how binary conversion errors impact octal representation.
4.	A processing unit converts inputs from multiple number systems into binary, performs operations, and converts results back into original formats. Improper sequencing has caused incorrect outputs in previous tests. The system must ensure proper data flow and accuracy. Formulate a correct sequence for number system conversion and infer the impact of incorrect sequencing.

5.	A system processes the decimal value 9.375 and converts it into binary and hexadecimal for computation and display. The conversion must follow positional representation rules without truncation. Determine binary and hexadecimal equivalents and justify each step logically.
6.	A computing system frequently converts large datasets between number systems. Binary ensures processing compatibility, while hexadecimal reduces storage size. Fractional conversions increase processing time. Assess trade-offs and recommend an efficient strategy for performance optimization.
7.	A communication system processes inputs in multiple number systems and converts them into binary for transmission. Outputs must be reconverted accurately with minimal overhead and high precision. Design a consistent model for number system conversion and justify its effectiveness.
8.	A digital calculator must represent negative numbers internally for performing arithmetic operations. The initial design used signed magnitude representation but faced inefficiencies in subtraction operations. The design team evaluates complement-based methods to improve performance and reduce hardware complexity. Determine the most appropriate representation method and justify the choice based on computational efficiency.
9.	A program implementing 1's complement subtraction produces incorrect results due to improper handling of carry. The developer notices missing logic when the addition generates an extra carry bit. Identify the fault and determine its impact on subtraction accuracy.

10.	A control system using 2's complement arithmetic produces incorrect outputs when adding large negative numbers. The system fails to detect overflow conditions, leading to incorrect interpretations of results. Diagnose the issue and infer how overflow affects system reliability.
11.	A processor performs subtraction using complement methods but occasionally produces incorrect results due to improper sequencing of operations. The system must ensure correct complement formation and addition. Formulate the correct sequence for complement-based subtraction and infer the impact of incorrect sequencing.
12.	A system computes $(-4) + (-2)$ using 4-bit 2's complement representation and must verify the correctness of the result within range limits. Determine the result using 2's complement and justify the steps logically.
13.	A processor design compares signed magnitude, 1's complement, and 2's complement for implementing arithmetic operations efficiently at scale. Assess trade-offs and recommend the best representation method.
14.	A digital system integrates signed number representation into arithmetic operations and must ensure consistency and accuracy across all computations. Design a consistent model for signed number representation and justify its effectiveness.
15.	A communication system must select one signal from multiple inputs based on control signals for transmission. The designer considers using either individual logic gates or a dedicated combinational module. The system requires minimal delay and efficient hardware usage. Determine the appropriate module for signal selection and justify the decision based on efficiency and performance.

